
HiMAP: Hierarchical Multi-Fidelity Active Prompt Optimization for Cost-Aware and Fair Auto-Prompting

Anonymous Author(s)

Affiliation

Address

email

Abstract

Automatic prompt optimization promises to adapt black-box language models without fine-tuning, yet prevailing methods are query-hungry, sequential, and narrow: they learn only after full evaluations, retrain surrogates from scratch per task, and optimize accuracy alone despite deployment constraints on cost, latency, and fairness. We present HiMAP, a hierarchical multi-fidelity active prompt optimizer that addresses these gaps by: (1) probing candidate prompts with cheap partial signals from the scorer model (first-k tokens and log-probabilities), (2) reusing cross-task regularities via a meta-surrogate pre-trained on public prompt-search traces and fine-tuned online, and (3) conducting Pareto-aware acquisition over accuracy, monetary cost, and gender bias. At each iteration, an optimizer LLM proposes candidates; the surrogate predicts objective means and uncertainties conditioned on task and fidelity; only the most promising advance to full evaluation while weak ones are early-dropped. Fairness is tracked using a small counterfactual calibration set. We describe a comprehensive protocol on GSM8K, BBH-Hard, MMLU subsets, and WinoGender under budgets up to 1,000 full-cost equivalents against OPRO, EvoPrompt, SAPS, and Random. A smoke-test run validates the infrastructure under tight memory limits, yielding stable training (loss 0.011) and a proxy embedding statistic; full multi-objective results are deferred to the complete protocol.

1 Introduction

Large language models excel at in-context learning but are notoriously brittle to prompt wording, motivating automatic prompt optimization for black-box APIs Yang et al. [2023], Arora et al. [2022]. Recent strategies iterate between proposing prompts and evaluating them with the target model, sometimes using the LLM itself as the optimizer Yang et al. [2023], or learning neural surrogates and bandit policies to cut query costs Lin et al. [2023], Chen et al. [2023]. Others pursue discrete optimization over prompt tokens Wen et al. [2023] or curate robust prompt ensembles Arora et al. [2022]. Despite progress, three pragmatic obstacles remain. First, surrogate models typically learn only from completed full evaluations, ignoring cheap partial signals such as the first few tokens or early log-probabilities of the scorer. Second, surrogates are routinely retrained from scratch per task, overlooking cross-task regularities in what constitutes a good prompt and thus incurring high cold-start costs. Third, optimization is almost always single-objective (accuracy), whereas real deployments require trade-offs involving accuracy, monetary and energy cost, latency, and fairness-bias Ma et al. [2023].

We aim to optimize prompts under realistic constraints: tight query budgets, wall-clock latency dominated by long completions, and the need to jointly manage accuracy, cost, and fairness. We introduce HiMAP, a hierarchical multi-fidelity active learning framework for prompt optimization.

HiMAP defines three fidelities per candidate: L0 cheap heuristics (prompt length, lexical diversity, and patterns such as "Let's think step-by-step" Kojima et al. [2022]), L1 partial evaluation that requests only the first 30 tokens for a handful of examples while returning token log-probabilities, and L2 full evaluation on the optimization set. A single meta-surrogate, pre-trained on public prompt-search traces spanning GSM8K, BBH, MMLU, ARC and others from OPRO logs Yang et al. [2023], predicts three objectives with uncertainty-accuracy, cost, and gender bias-conditioned on the prompt text, a task embedding, and the fidelity level. A Pareto-aware acquisition rule allocates which prompts to probe at which fidelity and promotes only candidates likely to extend the Pareto frontier, while an early-drop rule prunes weak prompts rapidly to save both budget and time.

This problem is challenging because prompt search induces a high-dimensional, rugged landscape where small edits can cause large performance swings Arora et al. [2022]. Multi-fidelity learning must cope with distribution shift between partial and full responses. Cross-task transfer requires a surrogate expressive enough to capture complex prompt-task interactions, yet stable under scarce task-specific data Phang et al. [2022]. Multi-objective optimization must avoid collapsing fairness or cost into a scalar objective, which can hide regressions, and instead reason about Pareto improvements efficiently.

Our approach follows three principles. First, hierarchical multi-fidelity modeling extracts value from low-cost probes to reduce expensive full evaluations and wall-clock delays. Second, meta-learning across tasks amortizes knowledge of prompt regularities, lowering cold-start overhead. Third, Pareto-aware acquisition with calibrated uncertainty focuses evaluations where they most likely improve accuracy while controlling cost and bias.

We make the following contributions:

- **Multi-fidelity formulation:** We formulate prompt optimization as hierarchical multi-fidelity active learning with L0/L1/L2 fidelities, enabling early filtering via partial scorer signals and cheap heuristics.
- **Meta-surrogate:** We introduce a multi-task meta-surrogate that conditions on prompt text, a task embedding, and fidelity to predict accuracy, cost, and gender bias with uncertainty; it is pre-trained on public optimization traces and fine-tuned online Yang et al. [2023].
- **Pareto acquisition:** We design a Pareto-aware acquisition and early-drop policy that allocates fidelity per candidate by upper/lower confidence bounds, improving budget use and wall-clock latency.
- **Bias estimation:** We integrate bias estimation via counterfactual gender swaps as an explicit optimization objective, following fairness-guided prompting insights Ma et al. [2023].
- **Evaluation protocol:** We detail a comprehensive experimental protocol across GSM8K, BBH-Hard, MMLU-5 subject subsets, and WinoGender with budgets of 250/500/1,000 full-cost equivalents, baselines including OPRO, EvoPrompt, SAPS, and Random, and extensive ablations and robustness analyses.
- **Release:** We release PromptTrace-Zoo and training scripts to facilitate surrogate reuse.

We validate feasibility under strict resource limits with a smoke test: fine-tuning the MiniLM encoder on streamed GSM8K with a lightweight objective achieved stable optimization (training loss 0.011) and a test embedding norm of 1.284. This confirms the data, quantization, and training pipelines, though it does not evaluate HiMAP's full loop. The remainder of this paper specifies the framework and evaluation protocol; running the complete experiments is deferred to future work.

2 Related Work

Prompt optimization as black-box search. OPRO proposes using an LLM as an optimizer that iteratively generates and evaluates candidate prompts, demonstrating strong gains on GSM8K and Big-Bench Hard Yang et al. [2023]. While effective, OPRO treats evaluation as single-fidelity and single-objective. HiMAP complements OPRO by introducing multi-fidelity probes, a meta-surrogate trained across tasks, and explicit multi-objective reasoning over accuracy, cost, and fairness.

Bayesian and bandit instruction optimization. Neural bandits coupled with transformers replace Gaussian processes with neural surrogates and leverage pretrained representations for instruction

88 optimization, improving query efficiency on black-box APIs Lin et al. [2023]. InstructZero uses
 89 Bayesian optimization over soft prompts that are decoded by an open model into instructions for a
 90 target black-box LLM Chen et al. [2023]. HiMAP similarly uses a neural surrogate but differs by:
 91 (1) modeling multiple fidelity levels to learn from partial signals; (2) pre-training a single surrogate
 92 across tasks using PromptTrace-style logs from OPRO Yang et al. [2023]; and (3) optimizing a Pareto
 93 frontier over accuracy, cost, and bias rather than scalarizing a single objective.

94 Discrete prompt search and local structure. Gradient-based discrete optimization can discover
 95 effective hard prompts for text tasks Wen et al. [2023] and restrict search to compact subspaces
 96 for diffusion models Wang et al. [2024]. ZOPO emphasizes that well-performing local optima are
 97 prevalent and incorporates an NTK-derived Gaussian process to search them efficiently Hu et al.
 98 [2024], Jacot et al. [2018]. HiMAP focuses on decision-efficient exploration rather than text-gradient
 99 surrogates, using partial, low-fidelity signals and cross-task meta-transfer to prioritize promising
 100 regions.

101 Aggregation and robustness. AMA shows that aggregating multiple imperfect prompts with weak
 102 supervision can substantially improve accuracy and robustness across tasks Arora et al. [2022].
 103 Embroid smooths predictions without changing prompts, leveraging representation neighborhoods
 104 to correct errors Guha et al. [2023]. HiMAP is orthogonal to these aggregation strategies; its
 105 multi-fidelity filtering and uncertainty estimates could be combined with ensemble aggregation
 106 downstream.

107 Fairness-aware prompting. Fairness-guided prompting reveals correlations between predictive bias
 108 and overall quality and proposes bias-aware prompt selection Ma et al. [2023]. HiMAP incorporates
 109 fairness explicitly as an optimization objective via a bias gap measured on counterfactually swapped
 110 examples, integrating it into both the surrogate and acquisition.

111 Foundational views on prompting and transfer. Analyses of prompting and prefix-tuning clarify
 112 capabilities and limitations of context-based adaptation Petrov et al. [2023], while universal approxi-
 113 mation results show surprising expressivity with prefixing Petrov et al. [2024]. Hypermodel-based
 114 transfer offers an avenue for reusing knowledge across tasks Phang et al. [2022]. These insights
 115 motivate HiMAP’s cross-task meta-surrogate and fidelity-aware learning.

116 Positioning. Prior work addresses different axes-query efficiency Lin et al. [2023], Chen et al. [2023],
 117 discrete optimization Wen et al. [2023], Wang et al. [2024], robustness Arora et al. [2022], Guha
 118 et al. [2023], fairness Ma et al. [2023], and transfer Phang et al. [2022]. HiMAP unifies multi-
 119 fidelity evaluation, meta-surrogate transfer, and multi-objective acquisition in a single framework for
 120 black-box prompt optimization.

121 3 Background

122 We study black-box prompt optimization. A candidate prompt p conditions a scorer language model
 123 S to map an input x to an output $\hat{y}$; a task T provides a small optimization set for evaluation and a
 124 large held-out test set. We optimize over three objectives: accuracy $A(p)$ on the optimization set,
 125 cost $C(p)$ measured in normalized query-equivalent units, and bias gap $B(p)$ defined as the absolute
 126 difference in accuracy across counterfactual gender swaps on a tiny calibration set.

127 Problem setting. For each candidate p and fidelity level $l \in \{L0, L1, L2\}$:

- 128 • L0: heuristic features $h(p)$ at zero cost, including prompt length, lexical diversity, and regex
 129 matches for reasoning triggers such as "Let’s think step-by-step" Kojima et al. [2022].
- 130 • L1: partial evaluation at cost 0.1, requesting only the first 30 tokens (temperature 0) for 4
 131 optimization examples and returning token-level log-probabilities. We aggregate simple
 132 proxies such as early token likelihoods aligned to gold choices.
- 133 • L2: full evaluation at cost 1 over the complete optimization set. The overall budget is
 134 measured in L2-equivalent units, with L0 free and L1 charged 0.1.

135 The multi-objective goal is to maximize A while minimizing C and B , approximating the Pareto set
 136 in $(A \uparrow, C \downarrow, B \downarrow)$ under a fixed budget.

137 Surrogate modeling. We learn a fidelity-aware surrogate f_θ that predicts a distribution over objectives
 138 conditioned on (p, t, l) , where t is a task embedding computed as the mean sentence-BERT embedding

of optimization examples. The surrogate outputs per-objective means and uncertainties; these inform acquisition decisions via confidence bounds. Meta-pretraining on heterogeneous prompt-search traces from OPRO Yang et al. [2023] provides cross-task regularities, followed by online fine-tuning on new $(p, l, \text{observed objectives})$ triples.

Acquisition and early stopping. At each outer iteration, an optimizer LLM proposes a batch of candidates. A Pareto-aware policy, using upper confidence bounds for accuracy and lower confidence bounds for cost and bias, selects which candidates to probe at L0 or L1 and which to promote to L2. Early stopping drops prompts whose post-L1 accuracy UCB falls below the current 10th percentile of observed accuracies, reducing both budget usage and wall-clock latency.

Context and assumptions. We adopt GPT-3.5-turbo-1106 as the scorer (temperature 0) and GPT-3.5-turbo as the optimizer (temperature 1.2). Datasets include GSM8K for math reasoning, subsets of MMLU, and BBH-Hard, plus WinoGender for fairness. Our fairness metric follows prior work on bias-aware prompting Ma et al. [2023]. The multi-fidelity structure and early-drop mechanisms are motivated by the high cost and latency of long completions in black-box APIs and by the observation that early signals and patterns such as zero-shot CoT can be informative Kojima et al. [2022].

4 Method

HiMAP comprises a hierarchical evaluation scheme, a cross-task meta-surrogate with calibrated uncertainty, and a Pareto-aware acquisition mechanism that allocates fidelity per candidate.

4.1 Hierarchical multi-fidelity evaluation

For each prompt p :

- L0 screening computes $h(p)$: prompt length, lexical diversity, and presence of reasoning cues like "Let's think step-by-step" Kojima et al. [2022]. These features augment the surrogate's input at zero cost.
- L1 partial evaluation issues four scorer calls restricted to the first 30 tokens with temperature 0 and log-probabilities enabled. We derive low-cost proxies such as the log-probability of the correct choice letter at early positions and the agreement between initial beams. This provides a roughly $10\times$ cheaper signal than full evaluation.
- L2 full evaluation runs on the complete optimization set, producing accurate estimates of $A(p)$ and updates for $B(p)$ on the calibration set.

4.2 Meta-surrogate architecture and training

The surrogate f_θ ingests: (1) a frozen sentence-transformers/all-MiniLM-L6-v2 encoder representation of the prompt text (dimension 384), (2) a task embedding t (dimension 128) computed as the mean over optimization-example embeddings, and (3) a learned fidelity embedding of size 32. These are concatenated and passed through a lightweight feed-forward head with hidden size 256 that outputs six values: per-objective means and log-variances for $o \in \{A, C, B\}$. Uncertainties are $\sigma_o = \exp(0.5 \cdot \log \text{var}_o)$. Offline, we pre-train f_θ on approximately 40 public optimization traces covering GSM8K, BBH, MMLU, ARC and related tasks from OPRO Yang et al. [2023], exposing the model to diverse prompt-task interactions and fidelity signals. Online, we fine-tune θ with newly observed (p, l, o) triples as HiMAP explores a new task, using AdamW with learning rate $1e-4$ and batch size 32. To stabilize cost prediction as token usage drifts, we normalize cost via an exponential moving average with decay $\gamma = 0.97$.

4.3 Pareto-aware acquisition with confidence bounds

Each outer iteration samples $N = 48$ candidates from the optimizer LLM (temperature 1.2, top-p 0.95). For each candidate, the surrogate predicts means μ_o and standard deviations σ_o for $o \in \{A, C, B\}$. We compute confidence-adjusted scores: $\text{UCB}_A = \mu_A + \beta \sigma_A$, $\text{LCB}_C = \mu_C - \beta \sigma_C$, $\text{LCB}_B = \mu_B - \beta \sigma_B$ with $\beta = 0.99$. A candidate is prioritized if the vector $(\text{UCB}_A, -\text{LCB}_C, -\text{LCB}_B)$ could extend the current Pareto frontier; ties are broken by larger uncertainty volume $\sigma_A \sigma_C \sigma_B$ to encourage information gain. We send 24 candidates to L0/L1 probes

187 and promote only $k = 6$ to L2 based on this ranking. Early stopping discards any candidate whose
 188 post-L1 UCB_A falls below the running 10th percentile of observed accuracies, conserving budget
 189 and reducing latency.

190 4.4 Fairness estimation and integration

191 For bias, we construct a tiny calibration set of 8 counterfactually swapped instances per task and
 192 define $B(p)$ as the absolute difference in accuracy across genders. We update $B(p)$ at L1 and L2 and
 193 feed these measurements into the surrogate and acquisition, following fairness-guided prompting that
 194 links predictive bias with degraded quality Ma et al. [2023].

195 4.5 Optimizer compatibility and transfer

196 HiMAP’s outer loop is agnostic to the proposal mechanism: OPRO-style iterative prompting Yang
 197 et al. [2023], neural bandits Lin et al. [2023], or discrete token edits Wen et al. [2023] can generate
 198 candidate prompts. The meta-surrogate amortizes cross-task knowledge akin to hypermodels Phang
 199 et al. [2022], lowering cold-start sample complexity.

200 4.6 Energy, cost, and latency accounting

201 We estimate API energy as $E = 0.0002$ kWh per 1,000 tokens plus 0.05 kWh per request, and
 202 carbon as $CO_2e = 417$ g/kWh $\times E$. We log surrogate FLOPs with fvcare and wall-clock latency
 203 and memory via standard profilers. These measurements both inform reporting and serve as training
 204 signals for the cost objective.

205 4.7 Implementation notes

206 We quantize all transformer encoders to 8-bit to minimize memory, keep surrogate peak GPU usage
 207 under 350 MB, and stream datasets to remain within a 500 MB system RAM cap. Hyperparameters
 208 are tuned once on GSM8K-dev and then frozen across tasks to avoid adaptive overfitting.

Algorithm 1 HiMAP outer loop with multi-fidelity acquisition

```

1: inputs: scorer LLM  $S$ , optimizer LLM  $O$ , meta-surrogate  $f_\theta$ , budget  $B$  in L2-equivalents
2: initialize observed set  $\mathcal{D} \leftarrow \emptyset$ , Pareto frontier  $\mathcal{P} \leftarrow \emptyset$ 
3: while  $B > 0$  do
4:   sample  $N$  candidate prompts  $\{p_i\}_{i=1}^N \sim O$ 
5:   for each  $p_i$  do
6:     compute L0 features  $h(p_i)$  and predict  $f_\theta(p_i, t, L0)$ 
7:   end for
8:   rank candidates by potential to extend  $\mathcal{P}$  using  $UCB_A, LCB_C, LCB_B$ 
9:   select top  $M$  for L1; for each, query partial outputs and update predictions
10:  early-drop any  $p$  with post-L1  $UCB_A$  below running 10th percentile
11:  promote top  $k$  survivors to L2; fully evaluate to obtain  $A(p), C(p), B(p)$ 
12:  update  $\mathcal{D} \leftarrow \mathcal{D} \cup \{(p, l, A, C, B)\}$  and fine-tune  $f_\theta$  on  $\mathcal{D}$ 
13:  update Pareto frontier  $\mathcal{P}$  with new  $(A, C, B)$ 
14:  decrement budget:  $B \leftarrow B - (k \cdot 1 + M' \cdot 0.1)$ , where  $M'$  is number probed at L1
15: end while
16: return Pareto frontier  $\mathcal{P}$  and best prompt(s)

```

209 5 Experimental Setup

210 5.1 Benchmarks and datasets

211 We evaluate on four public corpora hosted on HuggingFace Datasets: GSM8K v1 (grade-school math
 212 reasoning), BBH-Hard (the 38 hardest Big-Bench-Hard tasks), MMLU-5SUB (physics, computer
 213 science, philosophy, history, finance; 500 dev and 5,000 test questions total), and WinoGender-Hard
 214 (1,946 pronominal coreference pairs with gender swaps). Pre-processing follows task conventions:

215 GSM8K and BBH-Hard use author-provided inputs and answers; MMLU-5SUB strips option text
216 and converts to choice-letter format; WinoGender-Hard is prefixed with "Answer exactly either 'A'
217 or 'B'." Tokenization occurs inside GPT-3.5-turbo.

218 5.2 Splits and evaluation protocol

219 For each task, the optimization set contains 64 examples drawn from the development split and is
220 shared across methods and seeds. A 16-example validation set per task is reserved for early stopping
221 and surrogate diagnostics. Final evaluation uses the full official test sets, never exposed to optimizers.
222 The scorer LLM is GPT-3.5-turbo-1106 at temperature 0; the optimizer LLM is GPT-3.5-turbo at
223 temperature 1.2, top-p 0.95, generating 48 candidates per outer step.

224 5.3 Budgets and fidelities

225 We consider budgets of 250, 500, and 1,000 L2-equivalent units. Fidelity costs are $L_0 = 0$, $L_1 = 0.1$,
226 $L_2 = 1$. L_1 uses 30 tokens across 4 optimization examples per candidate. Per iteration, 24 candidates
227 undergo L_0/L_1 probing, with $k = 6$ promoted to L_2 based on Pareto-aware acquisition and early-drop
228 rules.

229 5.4 Surrogate and baselines

230 HiMAP’s surrogate uses sentence-transformers/all-MiniLM-L6-v2 in 8-bit (≈ 46 M parameters). An
231 encoder-swap ablation uses DistilBERT-base (≈ 66 M). Baselines include: Vanilla OPRO Yang et al.
232 [2023], EvoPrompt, SAPS (re-implemented with a DistilBERT-base surrogate trained from scratch
233 per task), and Random search. The surrogate is pre-trained offline on OPRO-style optimization traces
234 spanning GSM8K, BBH, MMLU, ARC Yang et al. [2023], then fine-tuned online on $(p, l, \text{objective})$
235 triples.

236 5.5 Hyperparameters and sensitivity

237 Unless stated otherwise: learning rate $1e-4$ (AdamW), batch size 32, $\beta_{UCB} = 0.99$, EMA $\gamma = 0.97$
238 for cost normalization. A sensitivity grid on GSM8K at budget 500 varies $\beta_{UCB} \in \{0.5, 1.0, 2.0\}$,
239 $L_1 \text{ tokens} \in \{15, 30, 60\}$, and $k_{\text{grad}} \in \{3, 6, 12\}$. We report median and 95% confidence intervals
240 across 5 random seeds for each method, budget, and task.

241 5.6 Metrics

242 Primary: 3-D Pareto hypervolume over normalized objectives with ideal $(1, 0, 0)$. Secondary: best
243 accuracy at fixed cost, bias gap at fixed accuracy, energy (kWh) and CO₂e using the API energy
244 model, and wall-clock time. As a diagnostic, we compute expected calibration error on GSM8K
245 log-probabilities.

246 5.7 Robustness analyses on GSM8K

247 We assess: (1) noise injection by permuting 20% of optimization example order; (2) distribution shift
248 by swapping USD/EUR tokens; (3) adversarial typos (two per question); and (4) domain transfer by
249 testing the final prompt on SVAMP.

250 5.8 Compute environment and logging

251 Experiments run on a single NVIDIA Tesla T4 (16 GB, CUDA 12) with a 500 MB system RAM cap.
252 Datasets are streamed with IterableDataset and mmap to keep RAM below 100 MB. All encoders
253 use 8-bit quantization (bitsandbytes), keeping surrogate peak GPU memory under 350 MB. OpenAI
254 API calls execute asynchronously on CPU; results are moved to GPU only for surrogate fine-tuning.
255 FLOPs are measured with fvcare; latency and memory via tracemalloc and nvidia-smi. Energy is
256 computed as $E = 0.0002 \text{ kWh per 1K tokens plus } 0.05 \text{ kWh per request, with CO}_2\text{e} = 417 \text{ g/kWh}$.

257 5.9 Ablations

258 We ablate: meta-pretraining (scratch initialization), partial evaluation (NoL1), acquisition strategy
259 (replace UCB with greedy top-k), and surrogate encoder (MiniLM vs DistilBERT-base). These isolate
260 contributions across algorithmic design, fidelity, and encoder choice.

261 5.10 What was executed for this submission

262 To validate the infrastructure under the stated constraints, we executed a smoke-test on GSM8K in
263 streaming mode using a small subset of 128 samples per split. We fine-tuned all-MiniLM-L6-v2 for
264 one epoch with a lightweight self-supervised objective (minimizing the L2 norm of the CLS embed-
265 ding) without invoking the scorer LLM and without hierarchical evaluation. The recorded outputs
266 were: `train_loss = 0.011175876409690986` and `test_embedding_norm = 1.2840004886489316`.
267 This run confirms the data and training pipelines but does not exercise HiMAP’s multi-fidelity,
268 multi-objective loop.

269 6 Results

270 We report only the outcomes of the smoke-test diagnostic described in the Experimental Setup.
271 The full HiMAP experiments with multi-fidelity evaluation, multi-objective optimization, baselines,
272 ablations, and robustness analyses were not executed for this submission. We therefore do not present
273 accuracy, cost, bias, or hypervolume results.

274 6.1 Smoke-test diagnostic on GSM8K (streaming subset)

- 275 • Setup: GSM8K in streaming mode with 128 samples per split; one-epoch fine-tuning of
276 sentence-transformers/all-MiniLM-L6-v2 under an L2-regularization proxy objective on the
277 CLS embedding; no scorer LLM calls; no L0/L1/L2 evaluation.
- 278 • Observed metrics:
 - 279 – `train_loss = 0.011175876409690986`
 - 280 – `test_embedding_norm = 1.2840004886489316`

281 6.2 Interpretation

282 The low training loss indicates stable optimization of the lightweight objective and validates the
283 execution scaffold: dataset streaming, 8-bit quantization, asynchronous logging, and fine-tuning
284 under strict GPU and RAM limits. The test embedding norm is a proxy statistic without a validated
285 link to downstream prompt-optimization objectives (accuracy, cost, bias); it should not be interpreted
286 as evidence for or against HiMAP’s effectiveness.

287 6.3 Limitations and fairness

288 No baselines were run, and none of HiMAP’s core components-hierarchical partial evaluation,
289 meta-surrogate pretraining and online adaptation, Pareto-aware acquisition, or bias estimation-were
290 exercised. Consequently, no claims about query efficiency, latency reduction, Pareto superiority, or
291 fairness can be made from this run.

292 6.4 Planned experiments

293 Once executed, we will report:

- 294 • Experiment 1: Query efficiency and multi-objective superiority on GSM8K, BBH-Hard,
295 MMLU-5SUB, and WinoGender-Hard at budgets 250/500/1,000 with 5 seeds, measuring
296 Pareto hypervolume, accuracy at fixed cost, bias gap at fixed accuracy, energy, carbon, and
297 wall-clock time, comparing against Vanilla OPRO Yang et al. [2023], EvoPrompt, SAPS,
298 and Random.
- 299 • Experiment 2: Meta-surrogate transfer on unseen tasks (ARC-Easy, StrategyQA, MMLU-
300 Law, MMLU-Astronomy), evaluating the number of L2 queries and wall-clock time to reach

301 a target accuracy and tracking surrogate MAE on a validation set; Wilcoxon signed-rank
 302 tests will assess significance; an encoder-swap replication will test robustness.

- 303 • Experiment 3: Latency benefits of multi-fidelity early stopping on GSM8K at budget 500,
 304 comparing HiMAP-Full (L0+L1+L2 with early-drop) to NoL1 and NoStop, targeting $\geq 2\times$
 305 speedup with ≤ 0.5 pp accuracy loss.

306 In summary, the present results confirm infrastructure readiness but do not evaluate HiMAP’s
 307 substantive contributions. Full experimental outcomes are deferred to future work.

308 7 Conclusion

309 We introduced HiMAP, a hierarchical multi-fidelity active framework for prompt optimization that
 310 jointly targets accuracy, cost, and fairness. HiMAP probes candidate prompts with inexpensive partial
 311 signals, transfers cross-task regularities via a pre-trained meta-surrogate, and allocates evaluations
 312 with a Pareto-aware acquisition rule that supports early dropping of weak candidates. A lightweight
 313 bias estimator based on counterfactual swaps integrates fairness directly into optimization, aligning
 314 the objective with real deployment needs Ma et al. [2023].

315 This submission reports a smoke-test diagnostic that validates the data pipeline, quantization, and fine-
 316 tuning under tight memory constraints, yielding stable training and a proxy embedding statistic. The
 317 core contributions-multi-fidelity evaluation, meta-surrogate transfer, and multi-objective acquisition-
 318 were not yet executed end-to-end; thus, we refrain from empirical claims regarding query efficiency,
 319 Pareto improvements, latency, or carbon reductions.

320 Future work will run the full protocol across GSM8K, BBH-Hard, MMLU-5SUB, and WinoGender-
 321 Hard under budgets up to 1,000, provide confidence intervals against OPRO, EvoPrompt, SAPS, and
 322 Random Yang et al. [2023], and include ablations on meta-pretraining, partial evaluation, acquisition
 323 strategy, and surrogate size Lin et al. [2023], Chen et al. [2023], Wen et al. [2023]. We further plan
 324 to extend HiMAP to RLHF-style reward tuning and to open-weight scorer models for fully local
 325 optimization. By unifying hierarchical signals, cross-task transfer, and multi-objective selection,
 326 HiMAP aims to make automatic prompt engineering more affordable, faster, and fairer in real-world
 327 deployments.

328 References

- 329 Simran Arora, Avanika Narayan, Mayee F. Chen, Laurel Orr, Neel Guha, Kush Bhatia, Ines Chami,
 330 Frederic Sala, and Christopher Ré. Ask me anything: A simple strategy for prompting language
 331 models. 2022.
- 332 Lichang Chen, Jiuhai Chen, Tom Goldstein, Heng Huang, and Tianyi Zhou. Instructzero: Efficient
 333 instruction optimization for black-box large language models. 2023.
- 334 Neel Guha, Mayee F. Chen, Kush Bhatia, Azalia Mirhoseini, Frederic Sala, and Christopher Ré.
 335 Embroid: Unsupervised prediction smoothing can improve few-shot classification. 2023.
- 336 Wenyang Hu, Yao Shu, Zongmin Yu, Zhaoxuan Wu, Xiangqiang Lin, Zhongxiang Dai, See-Kiong
 337 Ng, and Bryan Kian Hsiang Low. Localized zeroth-order prompt optimization. 2024.
- 338 Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and
 339 generalization in neural networks. *In Advances in neural information processing systems (pp.*
 340 *8571-8580) 2018*, 2018.
- 341 Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. Large
 342 language models are zero-shot reasoners. 2022.
- 343 Xiaoqiang Lin, Zhaoxuan Wu, Zhongxiang Dai, Wenyang Hu, Yao Shu, See-Kiong Ng, Patrick
 344 Jaillet, and Bryan Kian Hsiang Low. Use your instinct: Instruction optimization using neural
 345 bandits coupled with transformers. 2023.
- 346 Huan Ma, Changqing Zhang, Yatao Bian, Lemao Liu, Zhirui Zhang, Peilin Zhao, Shu Zhang, Huazhu
 347 Fu, Qinghua Hu, and Bingzhe Wu. Fairness-guided few-shot prompting for large language models.
 348 2023.

- 349 Aleksandar Petrov, Philip H. S. Torr, and Adel Bibi. When do prompting and prefix-tuning work? a
350 theory of capabilities and limitations. 2023.
- 351 Aleksandar Petrov, Philip H. S. Torr, and Adel Bibi. Prompting a pretrained transformer can be a
352 universal approximator. 2024.
- 353 Jason Phang, Yi Mao, Pengcheng He, and Weizhu Chen. Hypertuning: Toward adapting large
354 language models without back-propagation. 2022.
- 355 Ruochen Wang, Ting Liu, Cho-Jui Hsieh, and Boqing Gong. On discrete prompt optimization for
356 diffusion models. *Proceedings of the 41st International Conference on Machine Learning (ICML*
357 *2024)*, 2024.
- 358 Yuxin Wen, Neel Jain, John Kirchenbauer, Micah Goldblum, Jonas Geiping, and Tom Goldstein.
359 Hard prompts made easy: Gradient-based discrete optimization for prompt tuning and discovery.
360 2023.
- 361 Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V. Le, Denny Zhou, and Xinyun Chen.
362 Large language models as optimizers. 2023.